

PROJECT REPORT

OF

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2024

CERTIFICATE

I hereby certify that the work which is being presented in the project report entitled “**Smart Parking System**” by “Divyansh Verma, Jaya, Aryan Sharma” of B.Tech. Electronics & Communication Engineering submitted in the Department of Electronics & Communication Engineering submitted at **CHANDIGARH ENGINEERING COLLEGE, JHANJERI**, Mohali is an authentic record of my own work under the supervision of **Ms. RICHA MEHTA**.

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ABSTRACT

Parking management has become a critical challenge in urban areas due to the rapid increase in the number of vehicles and limited parking spaces. Traditional parking methods often lead to problems such as traffic congestion, unnecessary fuel consumption, and driver frustration caused by searching for vacant slots. These issues not only waste valuable time but also contribute to environmental pollution and reduced efficiency in transportation systems. Hence, there is a strong need for a modernized and automated solution to overcome these limitations.

This project proposes an **IoT-based Smart Parking System** that leverages sensor technology and cloud connectivity to provide real-time monitoring of parking slots. Ultrasonic and infrared sensors are used to detect the presence of vehicles in each slot, while a microcontroller processes the collected data. The information is then transmitted to a cloud server, which updates the availability status of parking spaces on a mobile application or web platform. This enables drivers to check slot availability before reaching the parking lot, thereby saving time and reducing congestion.

In addition to real-time slot detection, the system integrates a **servo motor-controlled barrier gate** to regulate entry and exit. This ensures secured access and prevents unauthorized usage of parking spaces. Furthermore, the system can be scaled to include features such as online reservation, digital payment, and data analytics to predict parking patterns. Such integrations enhance user convenience and help administrators manage parking facilities more efficiently.

The proposed IoT-based Smart Parking System not only addresses the challenges faced in traditional parking but also contributes to the development of **smart cities**. By minimizing search time, fuel consumption, and traffic congestion, the system promotes sustainable urban mobility. Its cost-effectiveness, scalability, and adaptability to existing infrastructures make it a practical and innovative solution for modern-day transportation needs.

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Chapter 1

INTRODUCTION

Parking has always been a basic requirement in cities, but with the rapid growth of urbanization, it has become a serious challenge. In the early days, when vehicles were fewer in number, small parking areas and manual systems were sufficient.

Attendants or guards were responsible for guiding drivers to available slots, and since traffic volumes were low, this approach was manageable. People rarely faced major issues with finding a place to park.

However, as cities expanded and industries developed, the number of vehicles increased at an unprecedented rate. Traditional manual systems soon proved inefficient and outdated.

Drivers in crowded urban centers began spending more and more time searching for available parking. This caused unnecessary delays in daily travel and created frustration for commuters.

The situation worsened in metropolitan regions where vehicle density is extremely high. Studies reveal that a large portion of traffic congestion in cities is caused by vehicles moving around only to find a parking space.

This constant circling not only wastes time but also consumes extra fuel. In turn, it increases transportation costs for individuals and contributes to noise and air pollution.

Environmental concerns add another dimension to the problem. Excess fuel burning releases harmful gases, while congested traffic leads to higher stress levels and reduced productivity in urban life.

These issues clearly show that conventional parking systems are no longer effective. A more efficient and intelligent system is required to handle the increasing number of vehicles and to make parking easier for drivers.

A **Smart Parking System** is designed to address these challenges by using technology to monitor slot availability in real time. It guides drivers directly to vacant spaces, helping reduce traffic congestion, save fuel, and improve the overall parking experience in cities.

What is IoT?

The **Internet of Things (IoT)** is one of the most transformative technologies of the modern era. At its core, IoT refers to the connection of physical devices — such as sensors, machines, and electronic systems — to the internet, allowing them to collect, exchange, and process data automatically.

In the past, computers and mobile phones were the primary devices connected to the internet. However, with IoT, everyday objects like appliances, vehicles, medical devices, and industrial machines can also be linked through a common network. This connectivity makes devices smarter and enables them to interact without requiring constant human intervention.

An IoT system typically consists of three main components: sensors or devices to gather data, a communication network to transmit this data, and a processing unit (cloud or local server) to analyze the information. The results are then used to trigger automated actions or to provide valuable insights to users.

For example, in agriculture, IoT devices monitor soil moisture and weather conditions to help farmers irrigate crops more efficiently. In healthcare, wearable IoT devices track heart rate and oxygen levels, alerting doctors in case of abnormalities. Such applications show how IoT improves decision-making and efficiency.

One of the key advantages of IoT is **real-time monitoring**. Devices can continuously observe changes in the environment and immediately share updates. This ensures quicker responses and better management of resources.

Another important feature of IoT is **automation**. By allowing devices to work together intelligently, IoT reduces the need for manual work. For instance, smart home systems can automatically turn off lights and appliances when they are not needed.

The relevance of IoT in transportation and parking systems is especially significant. By connecting sensors, controllers, and applications, IoT enables real-time tracking of vehicles, slot detection, and efficient traffic management.

Thus, IoT can be understood as a powerful network of “smart” devices that transforms ordinary systems into intelligent solutions. Its applications are wide-ranging and play a crucial role in building smarter, safer, and more sustainable cities.

What is Smart Parking System?

A **Smart Parking System** is an intelligent solution designed to address the growing challenges of parking management in urban areas. As cities expand and the number of vehicles increases, traditional parking methods—where drivers search manually for available slots—have become inefficient and time-consuming. Smart Parking Systems use modern technology to automate and optimize the process of finding, monitoring, and managing parking spaces.

The primary function of a Smart Parking System is **vehicle detection and slot monitoring**. Sensors such as infrared (IR) or ultrasonic devices are installed at parking slots to detect whether a space is occupied or free. This data is continuously monitored by a microcontroller or embedded system, which processes the information and updates the status in real time. This eliminates the need for human attendants to check parking availability, improving efficiency and reducing errors.

A key feature of smart parking systems is **real-time information sharing**. The system communicates parking availability to drivers through digital displays, mobile applications, or web interfaces. For example, a driver approaching a parking lot can instantly check which slots are free and be guided directly to an available space. This reduces the time spent searching for parking, decreases traffic congestion, and lowers fuel consumption, which in turn helps reduce environmental pollution.

Another important aspect of Smart Parking Systems is **automation of entry and exit**. Many systems integrate servo-controlled gates or automated barriers that open only when a valid payment or authentication is detected. Digital payment methods, such as QR codes or mobile transactions, can be integrated, eliminating the need for manual cash handling. This increases the convenience for users and ensures secure management of parking facilities.

IoT integration makes modern Smart Parking Systems even more efficient. By connecting sensors, controllers, and cloud platforms, IoT enables remote monitoring, real-time updates, and data analytics. For instance, parking lot administrators can track slot occupancy trends, identify peak hours, and optimize space utilization based on collected data. This capability also allows multiple parking lots to be monitored centrally, making large-scale urban management more effective.

Smart Parking Systems also contribute to **safety and security**. Automated gates prevent unauthorized access, and real-time monitoring ensures that parking areas are efficiently managed. Additionally, some systems can integrate cameras or alarms for added security, providing peace of mind to users.

Use of IoT in Smart Parking System

The **Internet of Things (IoT)** plays a vital role in making Smart Parking Systems efficient and intelligent. IoT connects sensors, microcontrollers, and cloud platforms, enabling real-time monitoring of parking slots. Each parking space is equipped with sensors that detect whether it is occupied or free. This information is processed by a microcontroller and transmitted to a cloud server or directly to a mobile application.

Through IoT, drivers can check parking availability before reaching the lot, reducing the time spent searching for free slots and minimizing traffic congestion. Mobile or web applications display the status of all parking slots, guiding users efficiently to empty spaces.

IoT also enables **automation of gates and payment systems**. Digital payments can be verified in real time, and servo-controlled gates open automatically for authorized vehicles. Additionally, IoT allows administrators to remotely monitor parking lot occupancy, analyze usage patterns, and optimize space allocation.

In short, IoT enhances Smart Parking Systems by enabling **real-time updates, automation, remote monitoring, and better resource management**, making parking faster, safer, and more convenient in modern urban areas.

Chapter 2

Methodology / Approach

The proposed **IoT-Based Smart Parking System** follows a systematic approach to monitor parking slots, guide users, and automate entry and exit. The methodology combines **sensor detection, microcontroller processing, IoT connectivity, and user interface management**.

1. Vehicle Detection:

The system uses an **ultrasonic sensor** installed at the entry gate to detect incoming vehicles. When a vehicle approaches, the sensor measures the distance and triggers the system to check slot availability. This eliminates the need for manual supervision at the entrance.

2. Slot Monitoring:

Each parking space is equipped with **infrared (IR) sensors** that detect whether a slot is occupied or vacant. The microcontroller continuously reads the sensor data and determines which slots are available.

3. Data Processing and IoT Integration:

The sensor data is processed by an **Arduino microcontroller** and sent to a cloud server or directly to a mobile application. IoT connectivity allows real-time updates on parking availability, which can be accessed remotely by users.

4. User Interaction:

The system displays slot status on a **16x2 LCD screen** and provides real-time updates via a **mobile app or web interface**. Drivers can quickly identify free slots without searching manually, saving time and reducing congestion.

5. Automated Gate and Payment:

For controlled entry, a **servo motor-operated barrier** is used. Payment verification can be done digitally through QR code scanning or IoT-enabled mobile payment. Once payment is confirmed, the gate opens automatically.

6. Continuous Monitoring and Updates:

The system operates continuously, providing real-time updates and ensuring efficient space utilization. Administrators can also monitor occupancy trends and optimize parking management using the IoT platform.

This **step-by-step methodology** ensures a fully automated, efficient, and user-friendly smart parking experience while reducing traffic congestion and improving urban parking management.

EQUIPMENT USED

This project was made using various electronic devices and sensor used for IoT technology. The names of main components are:

Arduino Uno

Breadboard

IR sensor

LCD Display

Ultra Sonic sensor

Servo Motor

Bluetooth Module (HC05)

Connecting Wires

3.1 Arduino Uno

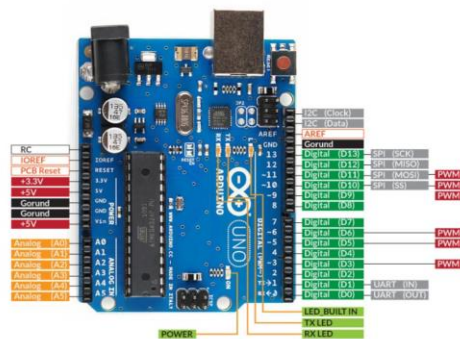


Figure 3.1: Arduino Uno

As seen in figure .1, Arduino **UNO** is one of the most popular and widely used **microcontroller development boards** in the Arduino family. It is designed to make electronics and programming easy for beginners, hobbyists, and even advanced users.

Features:

Microcontroller:

- ATmega328P (8-bit AVR RISC-based microcontroller).

Operating Voltage:

- 5V (logic level).

Input Voltage (recommended):

- 7–12V (through barrel jack or Vin pin).
- Can also work between 6–20V, but not recommended.

Digital I/O Pins:

- 14 total.
- 6 of them can be used as **PWM outputs** (pins 3, 5, 6, 9, 10, 11).

Analog Input Pins:

- 6 (A0–A5).
- Resolution: 10-bit (values from 0 to 1023).

Clock Speed:

- 16 MHz (with a quartz crystal).

Memory:

Flash Memory: 32 KB (0.5 KB used by bootloader).

SRAM: 2 KB.

EEPROM: 1 KB.

USB Connectivity:

- Standard Type-B USB for uploading code and serial communication.

Communication Protocols:

- UART (Serial), I2C, and SPI supported.

Current per I/O Pin:

- 20 mA (safe limit).
- Max 40 mA (absolute limit per pin).

Reset Button:

- For restarting the program easily.

Power Supply Options:

- USB cable.
- DC Barrel Jack (7–12V).
- Vin pin (external supply).

Built-in LED:

- Connected to pin 13 for quick testing.

Open-Source Hardware & Software:

- Easily modifiable and supported by a large community.

3.2 Breadboard

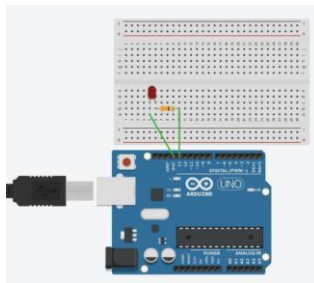


Figure 3.2: Breadboard

A breadboard is a tool used to build and test electronic circuits without soldering. It lets you connect electronic components like resistors, LEDs, sensors, microcontrollers (e.g., Arduino), and wires together temporarily.

Types of Breadboards

1. Mini Breadboard – small, portable, used for simple circuits.
2. Half-size Breadboard – medium size, good for small projects.
3. Full-size Breadboard – large, with many rows and columns, used for bigger circuits.

Structure of a Breadboard

- Power Rails (long side rows):
 - Usually marked with **red (+)** and **blue/black (-)** lines.
 - Used for supplying power (VCC and GND).
- Terminal Strips (main area in the center):
 - Divided into two halves by a **center gap** (called a trench).
 - Each vertical column of 5 holes is electrically connected.
 - The gap is used to place **ICs (Integrated Circuits)** so that each pin goes to a separate column.

3.3 IR Sensors

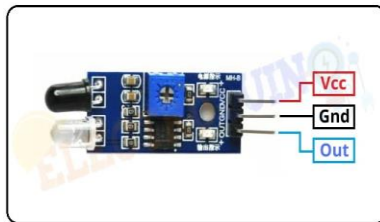


Figure 3.3: IR Sensor

An IR (Infrared) Sensor is an electronic device that uses infrared light to detect objects, measure distance, or sense motion.

- "Infrared" means light that is invisible to the human eye (wavelength longer than visible red light).
- IR sensors work by emitting infrared rays and then detecting how those rays are reflected, absorbed, or interrupted.

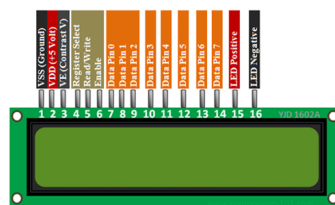
How does an IR Sensor work?

- The **emitter (IR LED)** continuously sends out IR light.
- If there is an object nearby:
- The IR light **reflects back** and is caught by the receiver.
- The receiver converts this into an **electrical signal**, which can be processed by a microcontroller (like Arduino UNO).

Applications of IR Sensors

- **Obstacle detection** (robots, vehicles, smart parking systems).
- **Line-following robots** (detecting black/white surfaces).
- **Remote controls** (TV, AC, etc. use IR communication).
- **Motion detection** (PIR sensors for security and lights).
- **Proximity sensing** (phones turning off screen near your ear).

3.4 LCD Display



*Figure 3.4: LCD 16*2*

An LCD (Liquid Crystal Display) is a flat-panel electronic display that uses liquid crystals and a backlight to show text, numbers, or graphics.

In electronics projects (like with Arduino UNO), the most common is the 16x2 LCD display.

- 16x2 means it can show 2 rows with 16 characters each (total 32 characters).
- Each character is displayed in a 5x8 dot matrix format.

How does an LCD work?

1. Liquid crystals don't emit light themselves but control how light passes through them.
2. A backlight (usually LED) shines behind the liquid crystals.
3. By applying voltage to the liquid crystals, they block or allow light, forming letters, numbers, or symbols.

Features of a 16x2 LCD

- Characters: 16 per row $\times$ 2 rows.
- Power Supply: 5V (compatible with Arduino UNO).
- Interface: Can be controlled in 4-bit or 8-bit mode.
- Pins: Usually 16 pins (for power, contrast, data, and control).
- Adjustable Contrast: Using a potentiometer (connected to V0 pin).
- Backlight: Helps in dark environments.

Common LCD Types in Projects

1. Character LCDs (like 16x2, 20x4) $\rightarrow$ show only text/numbers.
2. Graphical LCDs $\rightarrow$ display images/graphics.
3. TFT LCDs $\rightarrow$ full-color displays, like in phones.
4. LCD with I2C Module $\rightarrow$ reduces wiring (only 2 pins SDA & SCL needed).

Applications of LCD

- DIY projects (Arduino, Raspberry Pi).
- Digital meters (voltage, temperature, etc.).
- Clocks and counters.
- Home appliances (microwave ovens, washing machines).
- Industrial machines.

3.5 Ultra Sonic Sensor

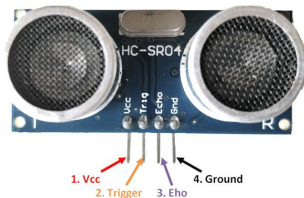


Figure 3.5: Ultra Sonic Sensor

An Ultrasonic Sensor is an electronic device that uses sound waves (at frequencies higher than humans can hear, i.e., >20 kHz) to measure distance or detect objects.

How does it work?

1. The sensor has two parts:
 - Transmitter (Trig pin): sends out ultrasonic sound waves.
 - Receiver (Echo pin): listens for the sound waves that bounce back from an object.
2. The sensor measures the time taken for the waves to return.
3. Since the speed of sound in air is known (~ 343 m/s), the distance can be calculated using:

Applications of Ultrasonic Sensor

- Obstacle detection in robots.
- Smart parking systems (detecting cars in slots).
- Automatic doors.
- Water level measurement in tanks.
- Blind-assistive devices.
- Industrial distance measurement tools.

3.6 Servo Motor



Figure 3.6: Servo Motor

A **servo motor** is a **small, precise, and controlled motor** used in robotics and automation to rotate or move an object to a specific angle.

Unlike normal DC motors (which keep rotating), a servo motor can be **commanded to rotate to a particular position (e.g., 0° , 90° , 180°)** and hold it there.

How does a Servo Motor work?

- Inside, it has:
 1. DC Motor – provides rotation.
 2. Gearbox – reduces speed & increases torque.
 3. Control Circuit – interprets signals.
 4. Feedback System (Potentiometer) – checks the current position.
- The servo receives a PWM (Pulse Width Modulation) signal from a microcontroller (like Arduino UNO).
- Based on the signal, it adjusts its shaft to the desired angle and locks it in place.

3.7 Bluetooth Module



Figure .7: Bluetooth Module (Hc-05)

A Bluetooth module is a small device that allows wireless communication between a microcontroller (like Arduino UNO) and other Bluetooth-enabled devices (like smartphones, laptops, or tablets).

It works on short-range radio frequency (2.4 GHz band) and is commonly used in IoT, robotics, and automation projects.

Most Common Bluetooth Module: HC-05 / HC-06

- HC-05 → can work in Master & Slave mode (two modules can talk to each other).
- HC-06 → works only in Slave mode (usually connected to a phone or laptop).

Features of Bluetooth Module (HC-05)

1. Operating Voltage: 3.6V – 6V (usually powered by Arduino 5V).
2. Communication Protocol: UART (TX & RX pins).
3. Frequency: 2.4 GHz ISM band.
4. Range: ~10 meters (Class 2).
5. Default Baud Rate: 9600 bps.
6. Modes:
 - Command Mode → configure name, password, baud rate.
 - Data Mode → send and receive data.

Pin Description (HC-05/06)

1. VCC → Power supply (5V).
2. GND → Ground.
3. TXD → Transmit Data (to Arduino RX).
4. RXD → Receive Data (from Arduino TX).
5. EN/KEY → Used to switch between Command & Data mode (only on HC-05).

How does it work?

1. Arduino sends/receives data through TX/RX pins.
2. Bluetooth module transmits this data wirelessly.
3. A mobile app (like Arduino Bluetooth Controller) or another Bluetooth device can receive it.

Applications of Bluetooth Module

- Wireless control of robots (via mobile app).
- Smart home automation (controlling lights, fans, appliances).
- Wireless data logging (temperature, humidity, etc.).
- Smart parking systems (mobile to Arduino communication).
- Remote-controlled vehicles/drones.

Chapter 4

Software and Program

The project is coded in Arduino C++ using Arduino IDE. This provides a simple environment for compiling and uploading code to the Arduino UNO.

Via Bluetooth module, user will receive UPI link in “Serial Bluetooth Terminal” app . After payment User have to type “Paid” to open gate.

Here’s a code of the project:

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <Servo.h>

#include <SoftwareSerial.h>


#define TRIG_PIN 9

#define ECHO_PIN 10

#define SERVO_PIN 6

#define BT_TX 11 // Arduino TX → HC-05 RX

#define BT_RX 12 // Arduino RX → HC-05 TX


// IR Sensors for Parking Slots

#define IR_SLOT_1 2

#define IR_SLOT_2 3
```

```
SoftwareSerial BT(BT_TX, BT_RX);
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
Servo gateServo;
```

```
void setup() {
```

```
    pinMode(TRIG_PIN, OUTPUT);
```

```
    pinMode(ECHO_PIN, INPUT);
```

```
    pinMode(IR_SLOT_1, INPUT);
```

```
    pinMode(IR_SLOT_2, INPUT);
```

```
    gateServo.attach(SERVO_PIN);
```

```
    lcd.init();
```

```
    lcd.backlight();
```

```
    BT.begin(9600);
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("Smart Parking");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("System Ready...");
```

```
    delay(2000);
```

```
}
```

```
void loop() {
```

```
    if (detectVehicle()) {
```

```
    if (checkParkingAvailability()) {  
        sendQRPayment();  
        waitForPayment();  
    } else {  
        lcd.clear();  
        lcd.setCursor(0, 0);  
        lcd.print("Parking Full!");  
        delay(3000);  
    }  
}  
}
```

// Detects if a vehicle is at the entry gate

```
bool detectVehicle() {  
    digitalWrite(TRIG_PIN, LOW);  
    delayMicroseconds(2);  
    digitalWrite(TRIG_PIN, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(TRIG_PIN, LOW);  
  
    long duration = pulseIn(ECHO_PIN, HIGH);  
    int distance = duration * 0.034 / 2;  
  
    return (distance < 10); // If a car is detected within 10 cm  
}
```

// Checks if any parking slots are available

```
bool checkParkingAvailability() {  
    int slot1 = digitalRead(IR_SLOT_1);  
    int slot2 = digitalRead(IR_SLOT_2);  
  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Slot Status:");  
  
    lcd.setCursor(0, 1);  
    lcd.print("S1:");  
    lcd.print(slot1 ? "Empty" : "Full");  
    lcd.setCursor(8, 1);  
    lcd.print("S2:");  
    lcd.print(slot2 ? "Empty" : "Full");  
  
    delay(2000);  
  
    return (slot1 || slot2); // If any slot is empty, return true  
}
```

// Sends QR Code Payment Link via Bluetooth

```
void sendQRPayment() {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Scan QR via BT");
```

```
    BT.println("Scan QR Code for Payment: https://upi_link"); // Replace with actual UPI link
```

```
}

// Waits for Payment Confirmation via Bluetooth
void waitForPayment() {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Waiting for Pay...");

    while (true) {
        if (BT.available()) {
            String receivedMessage = BT.readString();
            receivedMessage.trim();

            if (receivedMessage == "Paid") {
                lcd.clear();
                lcd.setCursor(0, 0);
                lcd.print("Payment Done!");
                lcd.setCursor(0, 1);
                lcd.print("Opening Gate...");

                openGate();
                break;
            }
        }
    }
}
```

```
// Opens the Gate

void openGate() {

    gateServo.write(90); // Open gate

    delay(5000);        // Keep open for 5 seconds

    gateServo.write(0); // Close gate


    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("Smart Parking");

    lcd.setCursor(0, 1);

    lcd.print("System Ready...");

}
```

Chapter 4

Conclusion/Future Scope

Conclusion

The IoT-based Smart Parking System demonstrates how embedded systems and IoT can be combined to solve real-world urban challenges. Parking management has always been a significant issue in cities due to the rapid growth of vehicles and limited space availability. The project addresses this challenge by introducing automation, digital monitoring, and user-friendly interaction. Through the integration of Arduino microcontroller, ultrasonic sensors, IR sensors, servo motors, LCD display, and Bluetooth communication, the system provides a low-cost and efficient solution for managing parking spaces.

The project successfully performs all its intended tasks. It detects the presence of a vehicle at the entrance using ultrasonic sensors, checks the availability of parking slots through IR sensors, and updates the status on an LCD screen for real-time monitoring. Furthermore, by adding a digital payment option via Bluetooth-based QR code, the system eliminates manual transactions and enhances user convenience. Once payment is confirmed, the servo-controlled gate operates automatically, making the entire process smooth and secure.

Another important conclusion from this project is the reliability and scalability of IoT-based systems. Even though the prototype works on a small scale, it can be implemented in larger parking lots or integrated into smart city infrastructure. The system reduces human intervention, minimizes errors, improves time efficiency, and enhances user experience. It also contributes indirectly to reducing traffic congestion and air pollution since drivers spend less time searching for parking.

Overall, the project proves that IoT-based smart systems are the future of urban infrastructure and can be implemented successfully with affordable electronic components.

Future Scope

Although the current system is functional and effective, there is significant potential for improvement and expansion. Future versions of the Smart Parking System can be made more advanced, intelligent, and user-oriented with the following enhancements:

1. **Mobile Application Integration** – Developing a smartphone app where users can view real-time parking availability, reserve slots in advance, and make payments securely.
2. **Cloud Connectivity** – Storing and analyzing parking data on a cloud server for centralized monitoring and predictive analytics, useful for both users and city administrators.
3. **Automatic Number Plate Recognition (ANPR)** – Using cameras and AI to automatically detect and register vehicles entering the parking area for enhanced security and tracking.
4. **RFID-based Entry/Exit** – Automating access for regular users with RFID tags or smart cards to reduce waiting time at gates.
5. **Solar-Powered Operation** – Making the system eco-friendly and sustainable by integrating renewable energy sources like solar panels.
6. **AI & Machine Learning** – Predicting peak parking times, suggesting optimal slot allocation, and improving traffic flow within parking areas.
7. **Smart City Integration** – Linking the system with city-wide IoT infrastructure to coordinate with traffic management systems, navigation apps, and smart payment gateways.

In the long run, such systems can evolve into fully automated parking solutions where human intervention is minimal. With proper development and deployment, this technology has the potential to revolutionize how urban parking is managed, reduce congestion, and contribute to sustainable urban living.

